

Room 1: Introduction to Cyber Security

5 Things I Learned

1. Introduction to cyber security is a general information about offensive security, defensive security and careers in cyber.
2. In offensive security, it teaches it is also called red team. The people who choose this field their work is to think like a hacker and search for vulnerabilities of the organization and company to prevent it from hack.
3. In defensive security, it teaches it is also called blue team. It's work to give protection to the company or organization. It has two responsibilities; 1. To prevent intrusions to enter the network. 2. To detect and protect the organization or company's network from intrusions.
4. In cyber careers, It gives brief information about all the roles and tells which roadmap to take in the specific field. It tells about security analyst, security engineer, digital forensic, incident responder, penetration tester, red teamer, malware analyst. Penetration tester and red teamer belong to offensive security, while most of the other fields like security analyst, incident responder and digital forensic belong to defensive security.
5. To conclude it, It gives basic understanding of all above fields and allow the person to decide in which field he/she interested or want to go in and also give rooms to start with.

Introduction to Cyber Security

Understand what is offensive and defensive security, and learn about careers available in cyber.

Explore both offensive and defensive security. Start by learning basic offensive security concepts, where you will hack a vulnerable online-banking application. Get exposure to defensive security and protect a system by blocking an ongoing cyber attack. In this module, you will also learn about the different careers within cyber security.



Offensive Security Intro

Hack your first website (legally in a safe environment) and experience an ethical hacker's job.



Defensive Security Intro

Introducing defensive security and related topics, such as Threat Intelligence, SOC, DFIR, Malware Analysis, and SIEM.



Careers in Cyber

Learn about the different careers in cyber security.



Topic Transition Recap

Questions from your recently completed rooms

Next steps



Introduction to Offensive Security

Understand what Offensive Security entails, and practice breaking into computer systems by exploiting applications and networks.



Introduction to Defensive Security

Learn Defensive Security by using digital forensics in an investigation and applying security operations to stop a live cyber attack.

Room 2: How the web works

5 Things I Learned

- 1.** The first room teaches about DNS which is Domain Name System. It tells how browser contact with webserver all over the world through internet. There is hierarchy level in it. root level, top level domain, second level domain and sub domain. There is two types of top level domain: 1. gTLD 2. ccTLD This is on right side of domain name. Sub domain is on left side of domain name. second level domain can use up to 63 characters; a to z, 0 to 9, -. It also tells we make so many sub domains for one website or app. There are also record types of DNS. A record, AAAA record, MX record, CName record, TXT record. It teaches what happens when we request to DNS. Computer first check to local cache this address is found locally or not. If not move to recursive DNS, it also searches in its local cache, if doesn't found there, it moves to root DNS. This redirect to real top level domain. The TLD moves towards authoritative server which is responsible for correct server and render the page browser request for. It stores and updates cache based on record types.
- 2.** The second room is about http in detail. Its full name is hypertext transfer protocol. It gives information about request and response, http methods, status codes, headers, cookies. In request and response, through url we give request about anything we search for or want. In response, it gives codes and tell whether request approve or not. In methods, we have get, post, put and delete methods which are used for view, add, update and delete. In status codes are codes like 200 for success, 404 page not found, 500 internal server error etc. In headers, it tells extra information we give and get in headers. They are used to save, update, redirect the extra information. In cookies, it is used for client side, it is used to store preferences of user and also we set cookies means customize it. What we want to see or search or what is our interest?
- 3.** The third room is how websites work, In it we request from browser which move to server through internet and server sends response back to browser through internet. Then, it tells about HTML which is structure of whole website. JAVASCRIPT is functional language and high level language, it is used to give functionality to the page. Also teach about the sensitive data exposure, in it developer forgets to clean the code or remove unnecessary data. this is vulnerability hacker used it to exploit the website. In html injection, the developer does not clean or filter the user input and hacker use it to embed the malicious code with it to hack the users.
- 4.** In the last room, we read about there is other components which helped in request and respond the website. Load balancers, content delivery networks, databases, web application firewall. Load balancer balances the load of servers and get request and give to the server which have less load so the response is quick. CDN helps in found the nearest physically located server and connects to them to cut off busy server all around the world. Database is used to store data online. WAF is used to protect data of organization or company. It allows or alert if see any suspicious activity.
- 5.** To conclude that, In how the web works, it tells us how the website, server, internet works behind the scenes which user can not see. How one request from user took place and get response. It walks through all the steps of website working, network and servers.

You earned a new badge!

Your dedication just paid off! You've earned a new badge.



World Wide Web

Completing the 'How The Web Works' module

Share your badge:

[in LinkedIn](#)[X Twitter / X](#)[f Facebook](#)[Go to link](#)

Share your win with your peers

Your progress can inspire others in your community.



Linux Fundamentals Part 1 completed!

sheikhamina037 completed another room on their cyber security journey.

Completed tasks

9

Points earned

88

Streak

3

Share on social media

Share your win before starting the next room

[in LinkedIn](#)[WhatsApp](#)[Telegram](#)[X Twitter / X](#)[f Facebook](#)[Reddit](#)[Go to dashboard →](#)

Room 3: Linux Fundamentals Part 1

5 Things I Learned

1. In linux fundamentals, we learn basics about linux. It is lightweight as compared to other operating systems. Linux powers the car entertainment / control panel, point of sale etc.
2. It gives few commands like echo and whoami; Echo gives output of any text we provide, Whoami used to find out username that we currently logged in.
3. It teaches some commands for file system like ls, cd, cat, pwd; ls is used for listing, cd is used for change directory, cat is used for concatenate, pwd is used for print working directory. If we search for some files we used word or command find, grep command is also used to search for content of files for specific value that we are looking for.
4. It also introduces to Shell operators: & is used to run commands in the background of your terminal. && is used to combine multiple commands together in one line of your terminal. > is redirector which is used to take command from terminal to somewhere else. >> it is same as > but it appends the output rather than replacing.
5. To conclude, It interacts with basic commands which are used in file system or must known.